SRI: N02LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 2$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*}} \left[\sum_{d=0}^{2} F_{s,d} + O(\mathrm{SR}^3)\right].$$

Relative order 0.

$$F_{s,0} = 1.0$$

Relative order 1.

$$F_{s,1} = 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1$$

Relative order 2.

$$\begin{aligned} F_{s,2} = {} & 1.5402686460338\epsilon_1^2 + 0.49988573877771\epsilon_2^2 \\ & + 1.0192767657904\epsilon_1\epsilon_2 + 0.14504832807052\epsilon_2\epsilon_3 \end{aligned}$$

1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2} \left[\sum_{d=0}^{2} F_{t,d} + O(\mathrm{SR}^3)\right].$$

Relative order 0.

$$F_{t,0} = 1.0$$

Relative order 1.

$$F_{t,1} = -0.54072569092296\epsilon_1$$

Relative order 2.

$$F_{t,2} = 1.5402686460338\epsilon_1^2 - 0.25062903478191\epsilon_1\epsilon_2$$

2 Consistency relations

2.1 $n = 0$

For $p = 2$, the absolute cutoff is $W = 3$:

$$\alpha_0^t = \sum_{d=1}^{3} R_{0,d} + O(\mathrm{SR}^4).$$

Absolute order 1.

$$R_{0,1} = -0.125r$$

Absolute order 2.

$$R_{0,2} = 0.0078125r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$\begin{aligned} R_{0,3} = &-0.0030712246746565r^3 + 0.125\alpha_1^s r \\ &- 0.10280837917801\left(\alpha_0^s\right)^2 r - 0.041327094794504\alpha_0^s r^2 \end{aligned}$$

2.2 $n = 1$

For $p = 2$, the absolute cutoff is $W = 4$:

$$\alpha_1^t = \sum_{d=2}^{4} R_{1,d} + O(\mathrm{SR}^5).$$

Absolute order 2.

$$R_{1,2} = 0.015625r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$\begin{aligned} R_{1,3} = &-0.0029296875r^3 + 0.125\alpha_1^s r \\ &- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2 \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{1,4} = &\, 0.0016576826498282r^4 + 0.125\alpha_2^s r \\ &+ 0.12398128438351\left(\alpha_0^s\right)^2 r^2 + 0.10280837917801\left(\alpha_0^s\right)^3 r \\ &+ 0.027641022071908\alpha_0^s r^3 - 0.072577094794504\alpha_1^s r^2 \\ &- 0.33061675835603\alpha_0^s \alpha_1^s r \end{aligned}$$

2.3 $n = 2$

For $p = 2$, the absolute cutoff is $W = 5$:

$$\alpha_2^t = \sum_{d=3}^{5} R_{2,d} + O(\mathrm{SR}^6).$$

Absolute order 3.

$$\begin{aligned} R_{2,3} = &-0.00390625r^3 + 0.125\alpha_1^s r \\ &- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2 \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{2,4} = &\, 0.0013427734375r^4 + 0.125\alpha_2^s r \\ &+ 0.125\left(\alpha_0^s\right)^3 r + 0.125\left(\alpha_0^s\right)^2 r^2 \\ &+ 0.025390625\alpha_0^s r^3 - 0.0625\alpha_1^s r^2 \\ &- 0.375\alpha_0^s \alpha_1^s r \end{aligned}$$

Absolute order 5.

$$\begin{aligned}R_{2,5} = &-0.00099348164677838r^5 + 0.125\alpha_3^s r \\ &+ 0.050668108270534\alpha_1^s r^3 - 0.13499234851023\left(\alpha_0^s\right)^2 r^3 \\ &- 0.28928966356152\left(\alpha_0^s\right)^3 r^2 - 0.020502543360439\alpha_0^s r^4 \\ &- 0.10280837917801\left(\alpha_0^s\right)^4 r - 0.33061675835603\left(\alpha_1^s\right)^2 r \\ &- 0.088202094794504\alpha_2^s r^2 + 0.63904189589007\left(\alpha_0^s\right)^2 \alpha_1^s r \\ &+ 0.46569385315053\alpha_0^s \alpha_1^s r^2 - 0.45561675835603\alpha_0^s \alpha_2^s r\end{aligned}$$

2.4 $n = 3$

For $p = 2$, the absolute cutoff is $W = 6$:

$$\alpha_3^t = \sum_{d=4}^{6} R_{3,d} + O(\mathrm{SR}^7).$$

Absolute order 4.

$$\begin{aligned}R_{3,4} = &\ 0.00146484375r^4 + 0.125\alpha_2^s r \\ &+ 0.125\left(\alpha_0^s\right)^3 r + 0.0234375\alpha_0^s r^3 \\ &+ 0.109375\left(\alpha_0^s\right)^2 r^2 - 0.0625\alpha_1^s r^2 \\ &- 0.375\alpha_0^s \alpha_1^s r\end{aligned}$$

Absolute order 5.

$$\begin{aligned}R_{3,5} = &-0.000762939453125r^5 + 0.125\alpha_3^s r \\ &+ 0.0419921875\alpha_1^s r^3 - 0.125\left(\alpha_0^s\right)^4 r \\ &- 0.375\left(\alpha_1^s\right)^2 r - 0.078125\alpha_2^s r^2 \\ &- 0.28125\left(\alpha_0^s\right)^3 r^2 - 0.01708984375\alpha_0^s r^4 \\ &- 0.1201171875\left(\alpha_0^s\right)^2 r^3 + 0.75\left(\alpha_0^s\right)^2 \alpha_1^s r \\ &+ 0.4375\alpha_0^s \alpha_1^s r^2 - 0.5\alpha_0^s \alpha_2^s r\end{aligned}$$

Absolute order 6.

$$\begin{aligned}R_{3,6} = &\ 0.0006988786133144 9r^6 + 0.125\alpha_4^s r \\ &+ 0.52264594794504\left(\alpha_1^s\right)^2 r^2 + 0.10280837917801\left(\alpha_0^s\right)^5 r \\ &+ 0.52433019631832\left(\alpha_0^s\right)^3 r^3 + 0.017257388611671\alpha_0^s r^5 \\ &+ 0.07369519446916\alpha_2^s r^3 + 0.1495700174229\left(\alpha_0^s\right)^2 r^4 \\ &+ 0.61990642191755\left(\alpha_0^s\right)^4 r^2 - 0.042328393296221\alpha_1^s r^4 \\ &- 0.1038270947945\alpha_3^s r^2 + 1.6087005501362\alpha_0^s\left(\alpha_1^s\right)^2 r \\ &+ 1.0946586542461\left(\alpha_0^s\right)^2 \alpha_2^s r + 0.71467513753404\alpha_0^s \alpha_2^s r^2 \\ &- 0.57385180946901\alpha_0^s \alpha_1^s r^3 - 1.9544879813691\left(\alpha_0^s\right)^2 \alpha_1^s r^2 \\ &- 1.1168502750681\alpha_1^s \alpha_2^s r - 1.0502754126021\left(\alpha_0^s\right)^3 \alpha_1^s r \\ &- 0.58061675835603\alpha_0^s \alpha_3^s r\end{aligned}$$